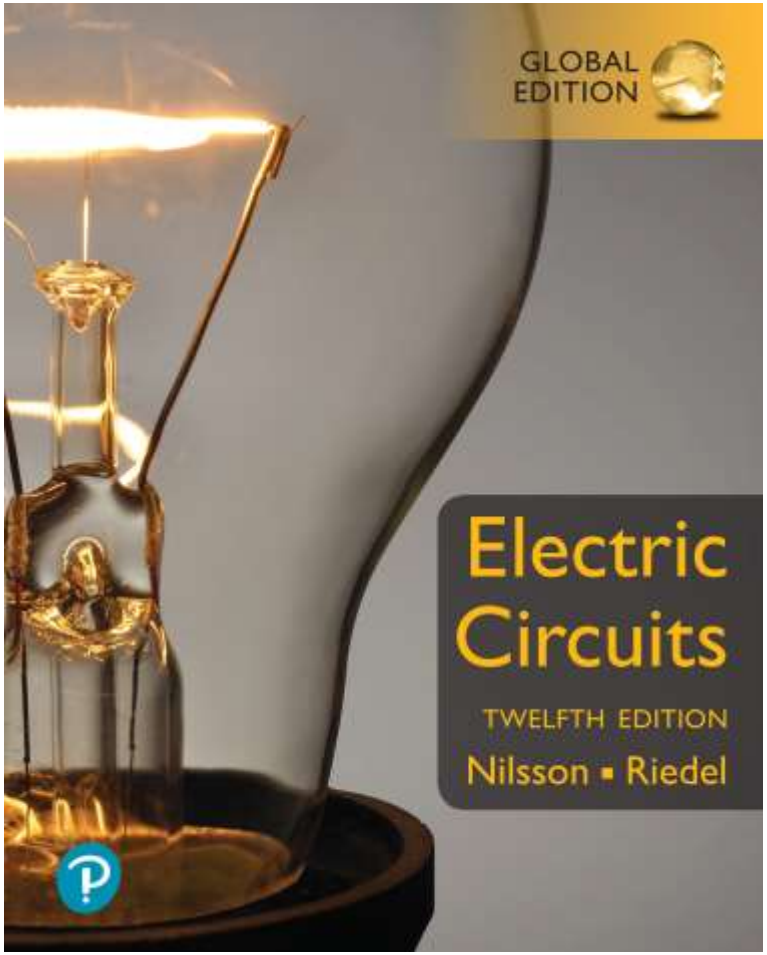
Electric Circuits

Twelfth Edition, Global Edition



Chapter 7 – Part 3

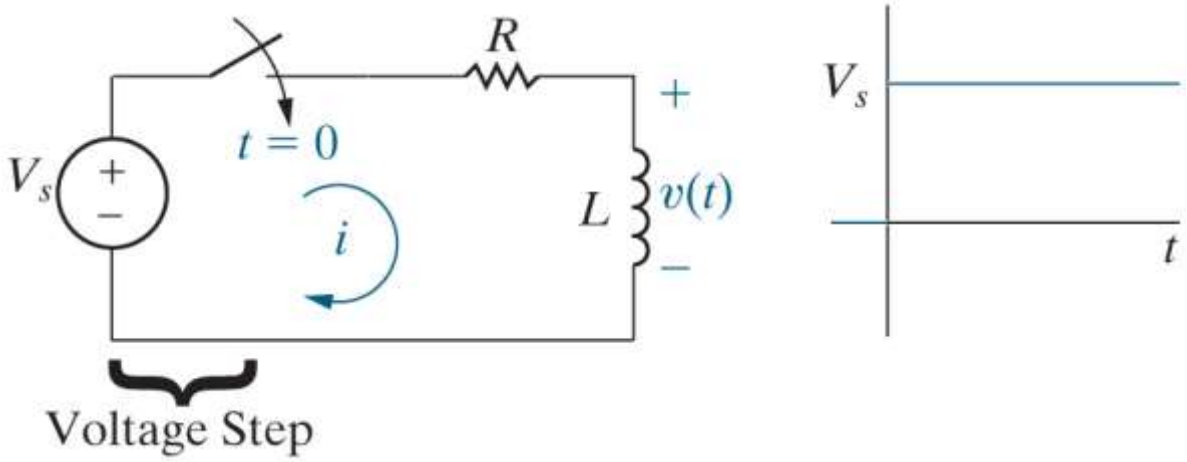
Response of First-Order *RL* and *RC* Circuits

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Chapter 7 – RL Step Response (1 of 10)

The problem: Given an initial current *i(0)* through *L* at $t = 0$, find *i(t)* for $t \geq 0$. Note that the initial current may be 0!

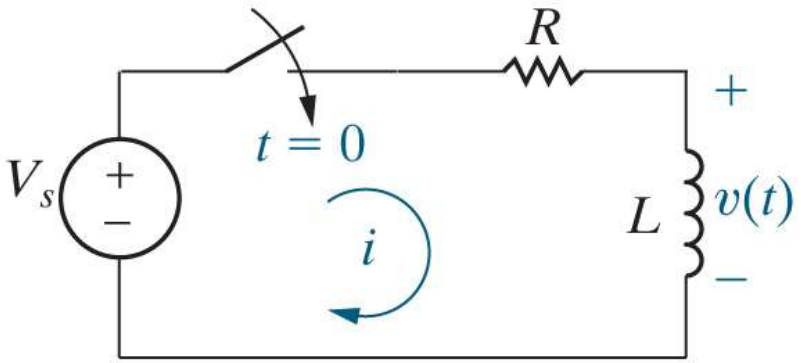


This is the step response of this circuit because for $t \geq 0$ there is an independent source in the circuit – the response is due to the initial energy stored in the inductor as well as the energy supplied by the voltage source for $t \geq 0$.

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Chapter 7 – RL Step Response (2 of 10)



KVL for $t \geq 0$:

$$-V_S + L\frac{di(t)}{dt} + Ri(t) = 0$$

$$\Rightarrow \quad L\frac{di(t)}{dt} + Ri(t) = V_S$$

This equation is a

- First-order (the highest derivative is a first derivative)
- Inhomogeneous (the right-hand side is non-zero)
- Ordinary differential equation
- With constant coefficients

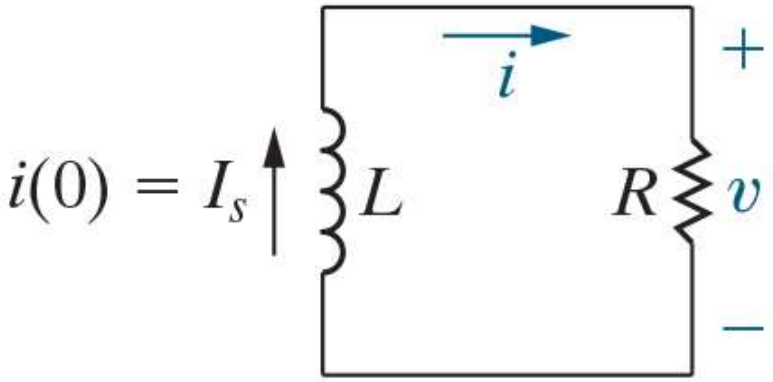
The left-hand side of the equation is identical to the describing differential equation for the *RL* Natural Response!

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Chapter 7 – RL Step Response (3 of 10)

The circuit we analyzed for the *RL* natural response problem is shown below, along with the expression for *i(t)*, for $t \geq 0$. What is the value of *i(t)* as $t \to \infty$?



$$i(t) = I_s e^{-(R/L)t}, \quad t \geq 0$$

A. *0*

B. I_s

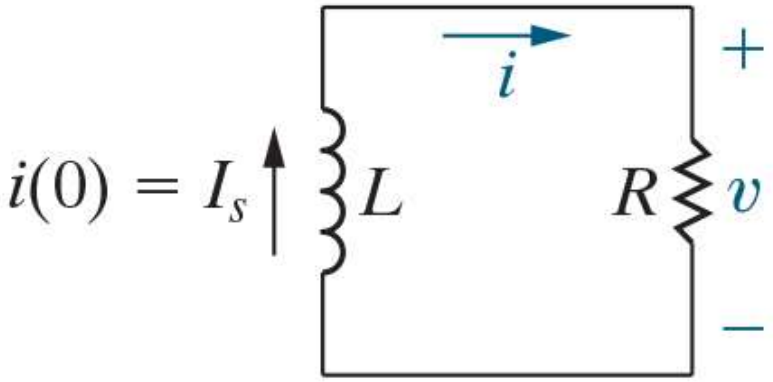
C. ∞

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Chapter 7 – RL Step Response (3 of 10)

The circuit we analyzed for the *RL* natural response problem is shown below, along with the expression for *i(t)*, for $t \geq 0$. What is the value of *i(t)* as $t \to \infty$?



$$i(t) = I_s e^{-(R/L)t}, \quad t \geq 0$$

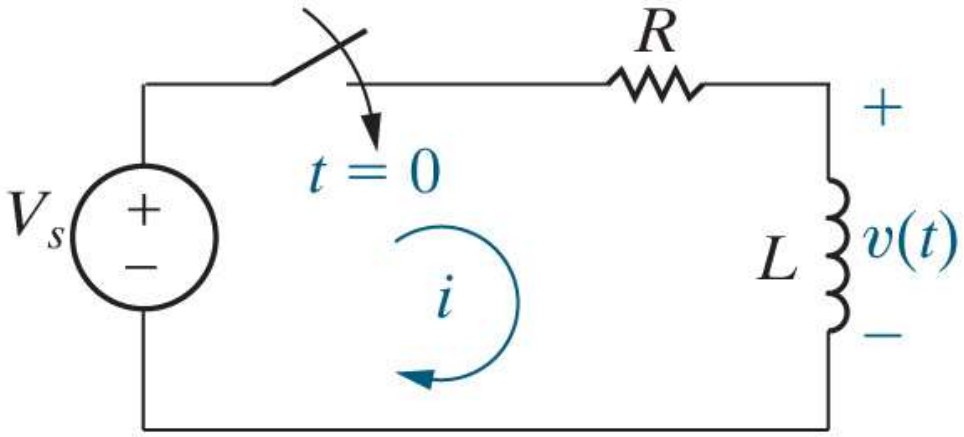
- ✔ A. *0*
- ✘ B. I_s
- ✘ C. ∞

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Chapter 7 – RL Step Response (4 of 10)

Now consider what happens to the current as $t \to \infty$ in the RL step response circuit. First, as $t \to \infty$, we can replace the inductor by



A. An open circuit

B. A short circuit

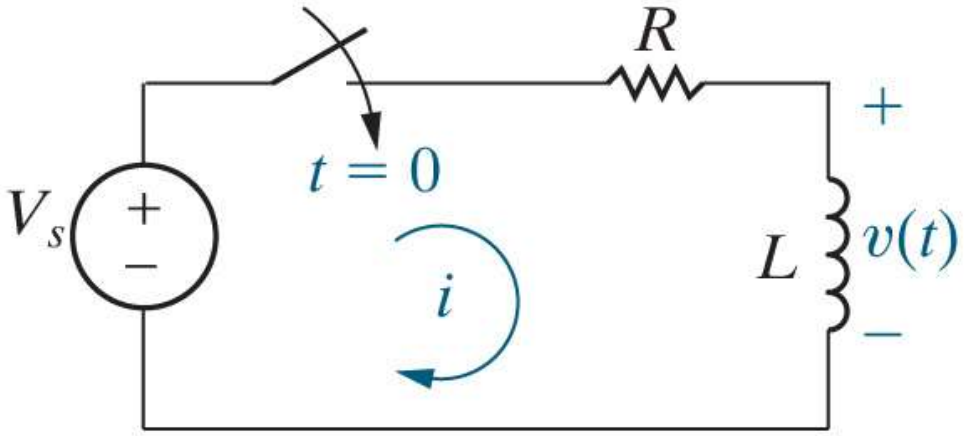
C. A resistor

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Chapter 7 – RL Step Response (4 of 10)

Now consider what happens to the current as $t \to \infty$ in the RL step response circuit. First, as $t \to \infty$, we can replace the inductor by



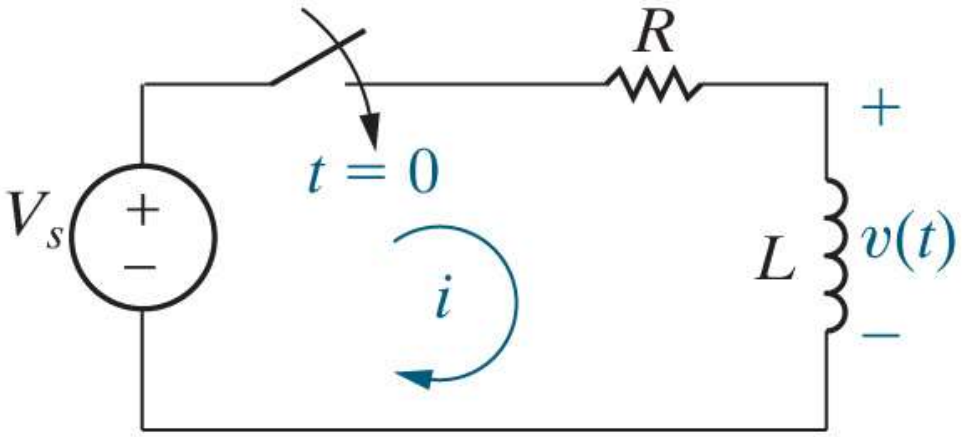
- ✗ A. An open circuit
- ✓ B. A short circuit
- ✗ C. A resistor

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Chapter 7 – RL Step Response (5 of 10)

To evaluate the current in the inductor as $t \to \infty$, close the switch and replace the inductor with a short circuit. The current in the short circuit is



A. 0

B. infinite

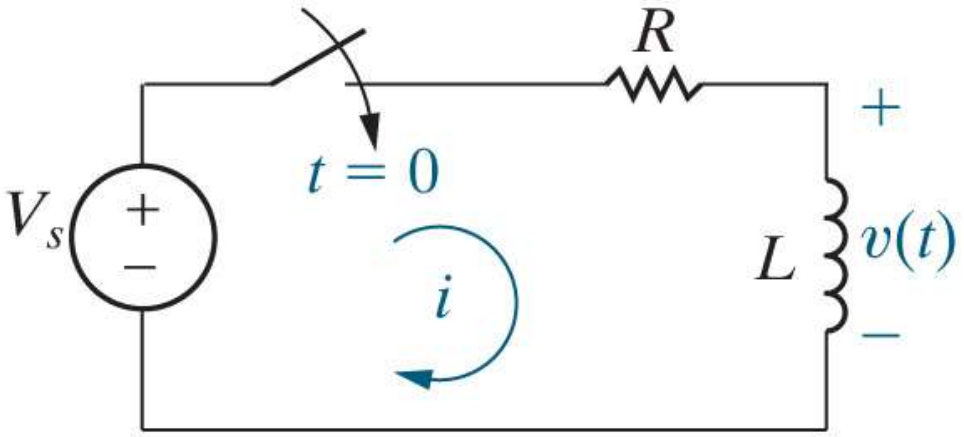
C. V_S / R

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Chapter 7 – RL Step Response (5 of 10)

To evaluate the current in the inductor as $t \to \infty$, close the switch and replace the inductor with a short circuit. The current in the short circuit is



✗ A. 0

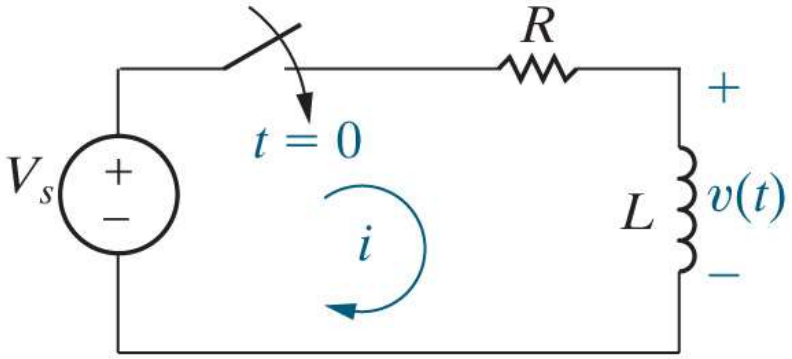
✗ B. infinite

✓ C. V_S / R

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Chapter 7 – RL Step Response (6 of 10)



KVL: $L\dfrac{di(t)}{dt} + Ri(t) = V_S$

The functional form for *i*(*t*) should be the same as for the natural response!

Try $i(t) = I_0 e^{-(R/L)t}$, where I_0 is the initial inductor current.

But as $t \to \infty$ $i(\infty) = I_0 e^{-(R/L)(\infty)} = 0$ which is incorrect!

Try $i(t) = I_F + I_0 e^{-(R/L)t}$,

where I_F is the inductor current as $t \to \infty$.

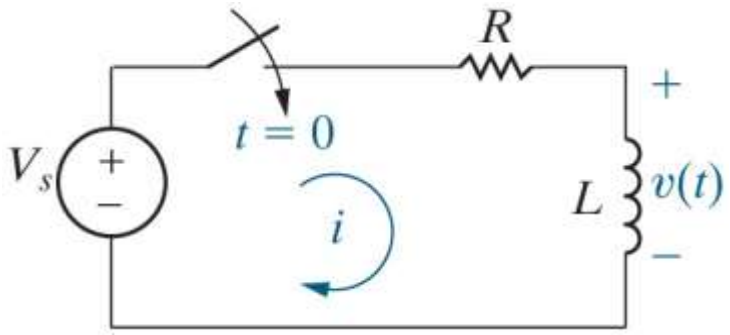
Then $i(\infty) = I_F + I_0 e^{-(R/L)(\infty)} = I_F$ (OK)

But $i(0) = I_F + I_0 e^{-(R/L)(0)} = I_F + I_0$ (NO!)

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Chapter 7 – RL Step Response (7 of 10)



KVL: $L\dfrac{di(t)}{dt} + Ri(t) = V_S$

Now try $i(t) = I_F + (I_0 - I_F)e^{-(R/L)t}$

Then $i(\infty) = I_F + (I_0 - I_F)e^{-(R/L)(\infty)} = I_F$ (OK)

And $i(0) = I_F + (I_0 - I_F)e^{-(R/L)(0)} = I_F + I_0 - I_F = I_0$ (OK)

Make sure that this *i*(*t*) satisfies the differential equation:

$$i(t) = I_F + (I_0 - I_F)e^{-(R/L)t}; \qquad \frac{di(t)}{dt} = -(R/L)(I_0 - I_F)e^{-(R/L)t}$$

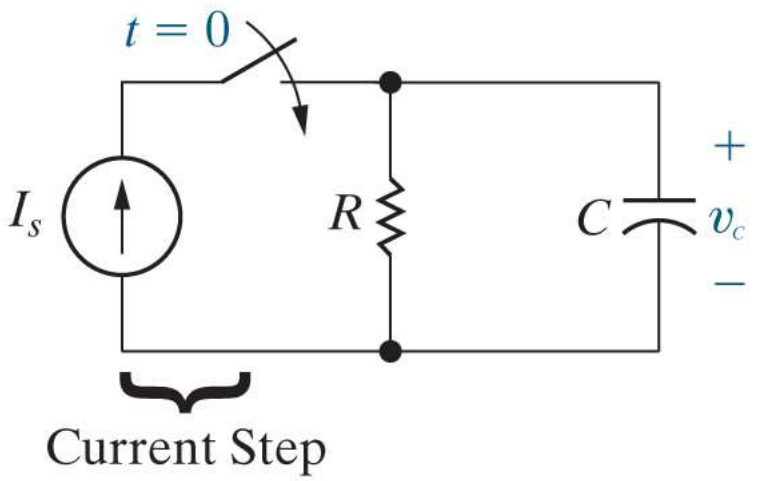
Substituting,

$$\mathrm{L}[-(R/L)(I_0 - I_F)e^{-(R/L)t}] + R[I_F + (I_0 - I_F)e^{-(R/L)t}] = RI_F = V_S$$

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Chapter 7 – RL Step Response (8 of 10)



The problem: Given an initial voltage drop *v(0)* across *C* at $t = 0$, find *v(t)* for $t \geq 0$. Note that the initial voltage may be 0!

This is the duel of the *RL* step response problem, so its solution is the duel of the solution to the *RL* step response!

$$v_C(t) = V_F + (V_0 - V_F)e^{-t/RC}, \qquad t \geq 0$$

where $V_0 = v_C(0)$ and $V_F = v_C(\infty)$

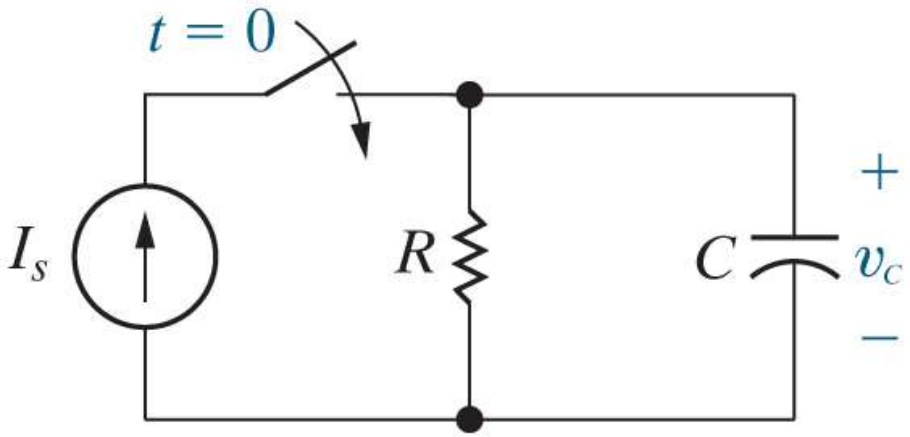
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Chapter 7 – RL Step Response (9 of 10)

Here is the circuit for the *RC* step response problem.

As $t \to \infty$, the capacitor can be replaced by



A. An open circuit

B. A short circuit

C. A constant current

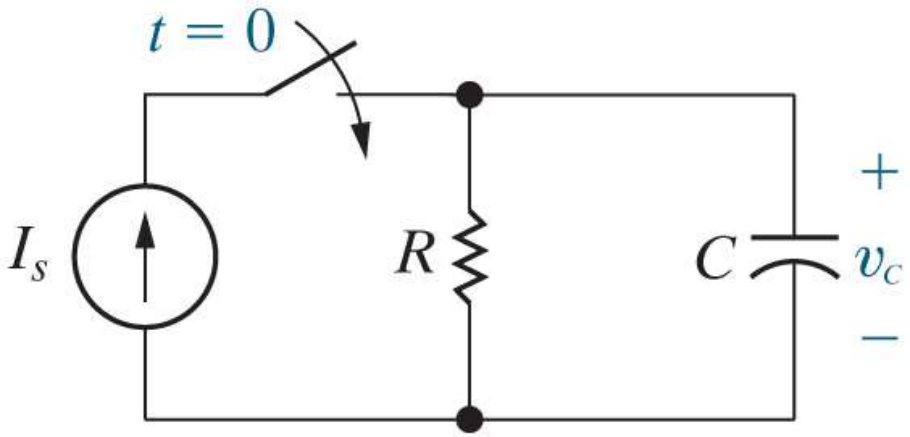
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Chapter 7 – RL Step Response (9 of 10)

Here is the circuit for the *RC* step response problem.

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✔ A. An open circuit

✘ B. A short circuit

✘ C. A constant current

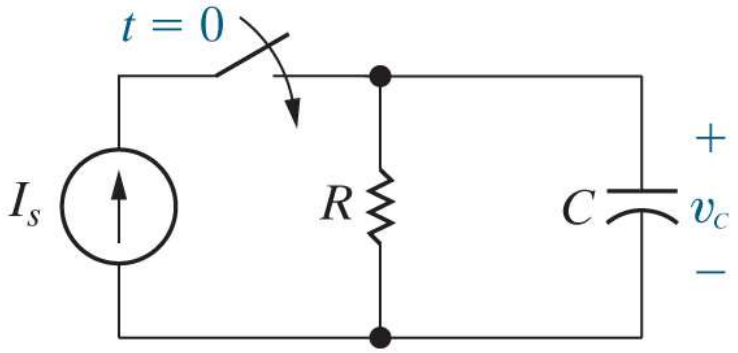
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Chapter 7 – RL Step Response (10 of 10)

To evaluate the voltage drop across the capacitor as $t \to \infty$,

close the switch and replace the capacitor with an open circuit. The voltage drop across the open circuit, positive at the top, is



A. 0

B. $I_S R$

C. infinite

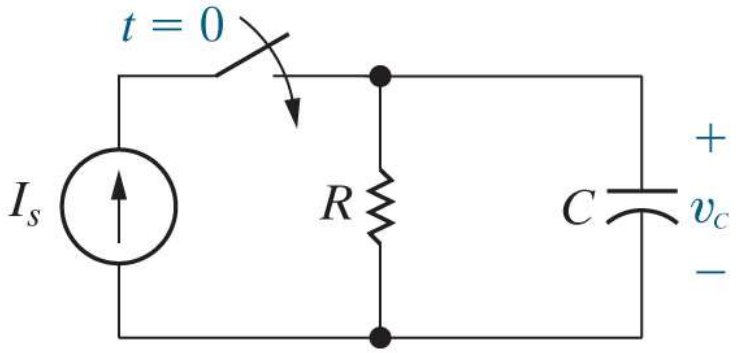
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Chapter 7 – RL Step Response (10 of 10)

To evaluate the voltage drop across the capacitor as $t \to \infty$,

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✗ A. 0

✓ B. $I_S R$

✗ C. infinite

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Chapter 7 – *RL*/*RC* Step Response

A general solution method:

Step 1 – Identify the variable $x(t)$, which is the inductor current for *RL* circuits and capacitor voltage for *RC* circuits.

Step 2 – Calculate the initial value X_0 by analyzing the circuit to find $x(t)$ for $t < 0$.

Step 3 – Calculate the time constant τ; for *RL* circuits, $\tau = R / L$ and for *RC* circuits, $\tau = RC$, where *R* is the equivalent resistance connected to the inductor or capacitor for $t \geq 0$.

Step 4 – Calculate the final value X_f by analyzing the circuit to find $x(t)$ as $t \to \infty$.

Step 5 – Write the equation for $x(t)$,

$$x(t) = X_f + (X_0 - X_f)e^{-t/\tau}, \text{ for } t \geq 0.$$

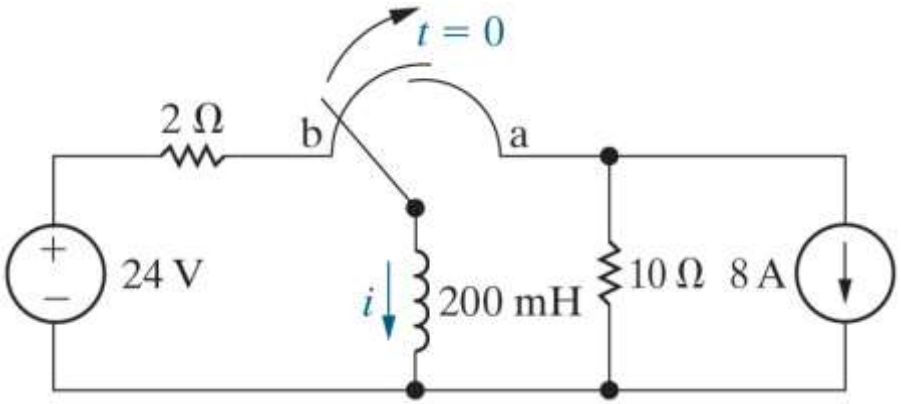
Step 6 – Calculate other quantities of interest using $x(t)$.

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Chapter 7 – *RL*/*RC* Step Response Examples (1 of 10)

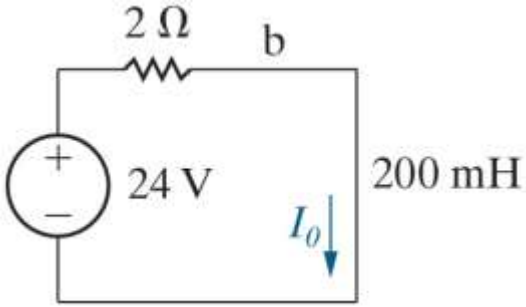
Find i(t) for $t \geq 0$.



Step 1 – Identify the variable $x(t)$, which is the inductor current labeled i in the circuit.

Step 2 – Calculate the initial value I_0 by analyzing the circuit to find $i(t)$ for $t < 0$.

For $t < 0$



$$I_0 = \frac{24}{2} = 12\,\text{A}$$

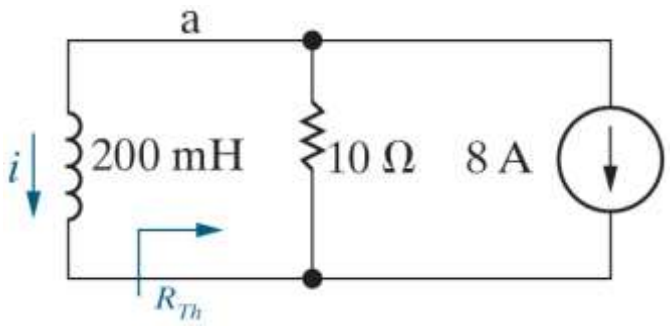
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Chapter 7 – *RL/RC* Step Response Examples (2 of 10)

Step 3 – Calculate the time constant, $\tau = \mathrm{L/R_{eq}}$, for $t \geq 0$.

What is the Thevenin equivalent resistance of the circuit attached to the inductor?



A. $10\,\Omega$

B. $80\,\Omega$

C. $200\,\Omega$

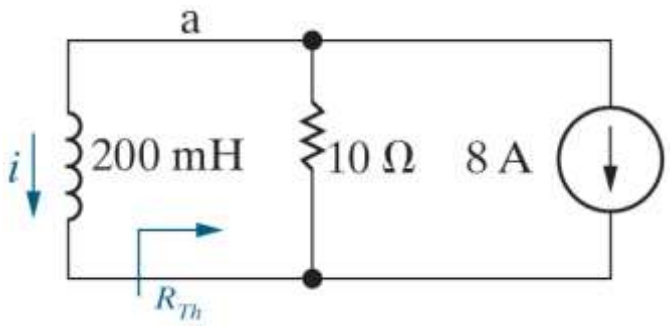
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Chapter 7 – *RL/RC* Step Response Examples (2 of 10)

Step 3 – Calculate the time constant, $\tau = \mathrm{L/R_{eq}}$, for $t \geq 0$.

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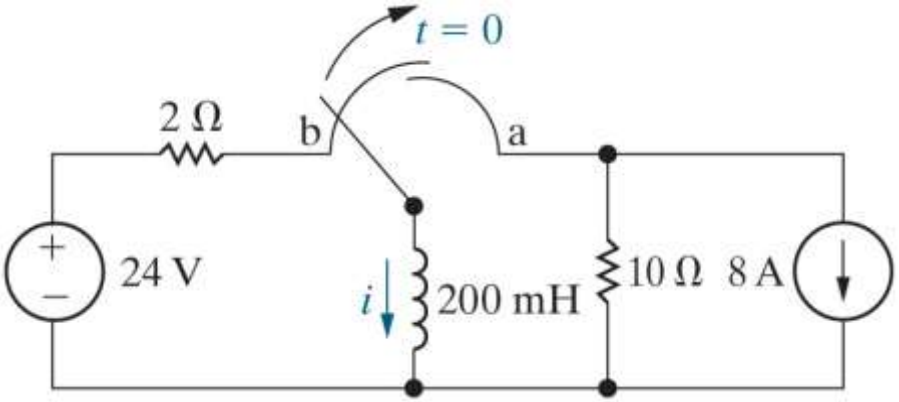
- ✔ A. $10\,\Omega$
- ✘ B. $80\,\Omega$
- ✘ C. $200\,\Omega$

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Chapter 7 – *RL*/*RC* Step Response Examples (3 of 10)

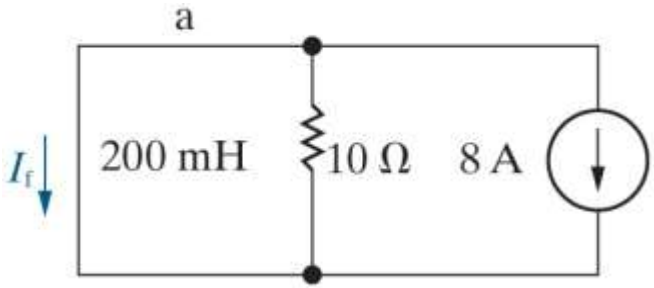
Find i(t) for $t \geq 0$ (continued).



The time constant is $\tau = R / L = (0.2) / (10) = 0.02$ s.

Step 4 – Calculate the final value I_f by analyzing the circuit to find $i(t)$ as $t \to \infty$.

As $t \to \infty$



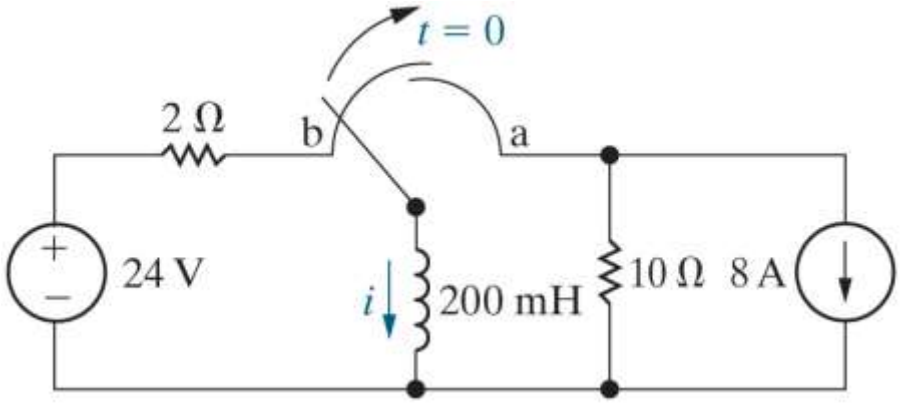
$$I_f = -8 \text{ A}$$

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Chapter 7 – *RL*/*RC* Step Response Examples (4 of 10)

Find i(t) for $t \geq 0$ (continued).



Step 5 – Write the expression for the inductor current:

$$i(t) = I_F + (I_0 - I_F)e^{-t/\tau}$$

$$= -8 + [12 - (-8)]e^{-t/0.02}$$

$$= -8 + 20e^{-50t} \text{ A}, \quad t \geq 0$$

Check: $i(0) = -8 + 20e^{-50(0)} = -8 + 20 = 12$ A (OK)

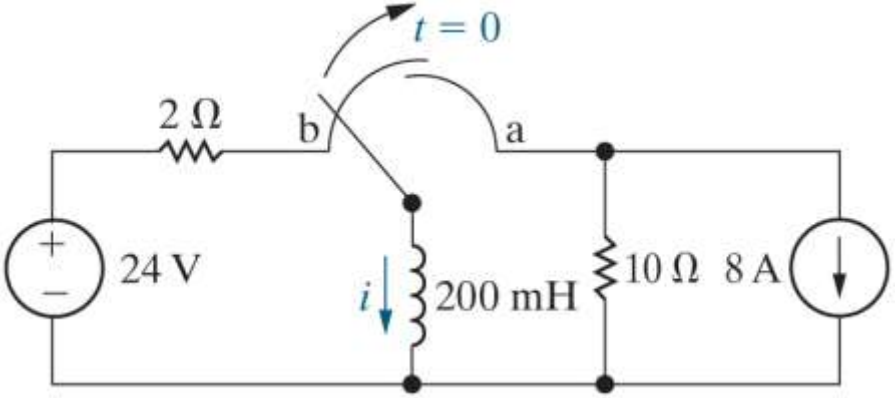
$i(\infty) = -8 + 20e^{-50(\infty)} = -8$ A (OK)

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Chapter 7 – *RL*/*RC* Step Response Examples (5 of 10)

Find the current in the resistor (flowing from top to bottom), given the inductor current just calculated, for $t \geq 0$.



$$i(t) = -8 + 20e^{-50t} \text{ A}, \quad t \geq 0$$

A. $-8 + 20e^{-50t}\,\text{A}, t \geq 0$

B. $20e^{-50t}\,\text{A}, t \geq 0$

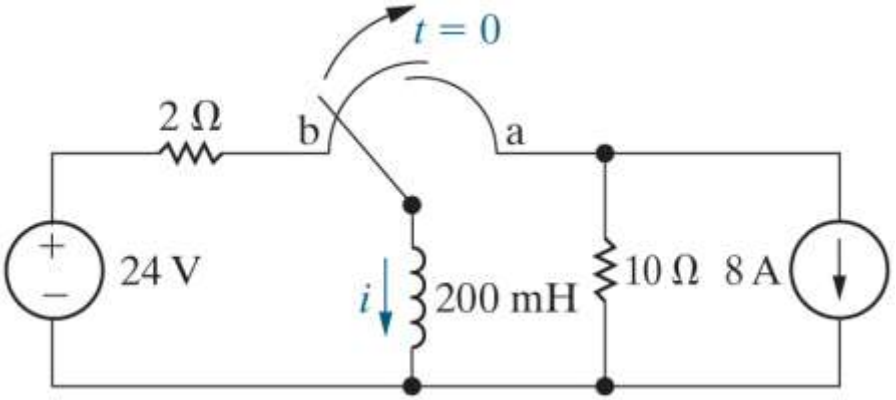
C. $-20e^{-50t}\,\text{A}, t \geq 0$

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Chapter 7 – *RL*/*RC* Step Response Examples (5 of 10)

Find the current in the resistor (flowing from top to bottom), given the inductor current just calculated, for $t \geq 0$.



$$i(t) = -8 + 20e^{-50t} \text{ A}, \quad t \geq 0$$

✗ A. $-8 + 20e^{-50t}\,\text{A}, t \geq 0$

✗ B. $20e^{-50t}\,\text{A}, t \geq 0$

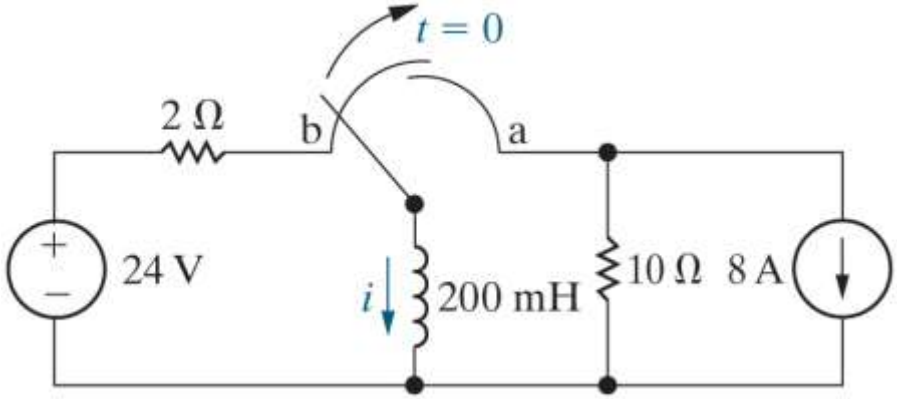
✓ C. $-20e^{-50t}\,\text{A}, t \geq 0$

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Chapter 7 – *RL*/*RC* Step Response Examples (6 of 10)

Find the voltage drop across the resistor (positive at the top), given the resistor current just calculated, for $t \geq 0^+$.



$$i_{10\Omega}(t) = -20e^{-50t} \text{ A}, \quad t \geq 0$$

A. $-2e^{-50t}$ V, $t \geq 0^+$

B. $200e^{-50t}$ V, $t \geq 0^+$

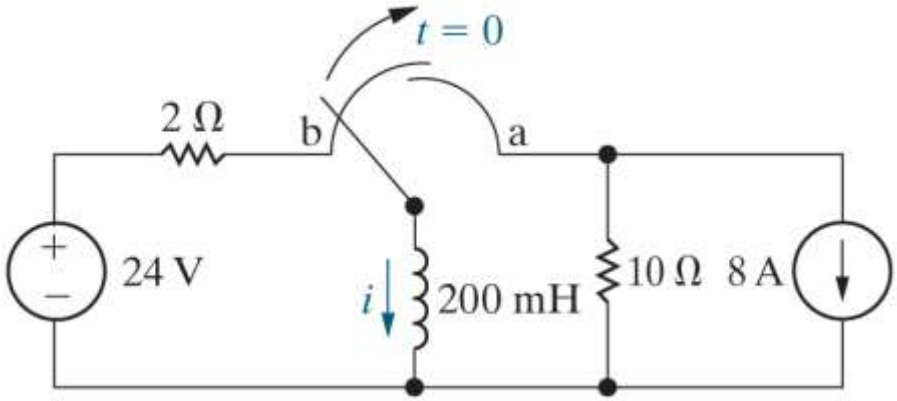
C. $-200e^{-50t}$ V, $t \geq 0^+$

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Chapter 7 – *RL*/*RC* Step Response Examples (6 of 10)

Find the voltage drop across the resistor (positive at the top), given the resistor current just calculated, for $t \geq 0^+$.



$$i_{10\Omega}(t) = -20e^{-50t} \text{ A}, \quad t \geq 0$$

✗ A. $-2e^{-50t}$ V, $t \geq 0^+$

✗ B. $200e^{-50t}$ V, $t \geq 0^+$

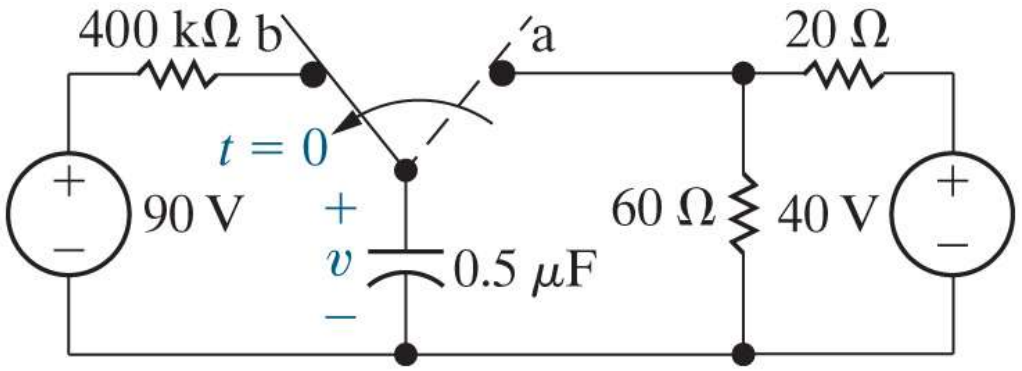
✓ C. $-200e^{-50t}$ V, $t \geq 0^+$

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Chapter 7 – *RL*/*RC* Step Response Examples (7 of 10)

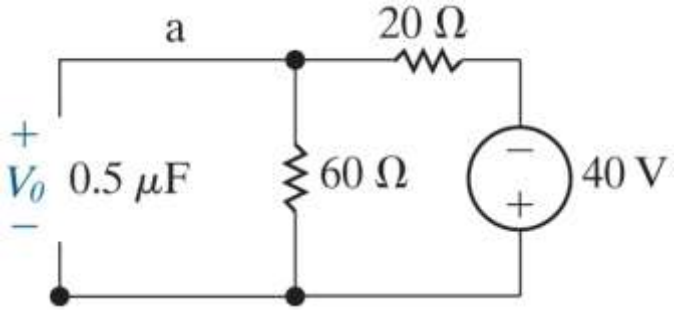
Find $v(t)$ for $t \geq 0$.



Step 1 – The variable $x(t)$ is the capacitor voltage drop, which is already defined in the circuit.

Step 2 – Calculate the initial value of the voltage drop across the capacitor, V_0.

For $t < 0$



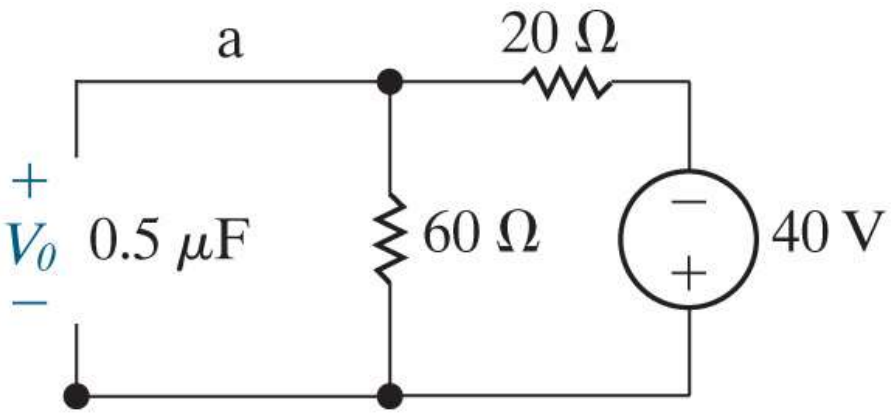
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Chapter 7 – *RL/RC* Step Response Examples (8 of 10)

What is the best circuit analysis tool to use when finding the initial voltage drop across the capacitor?

For $t < 0$



A. Current division

B. Voltage division

C. Source transform

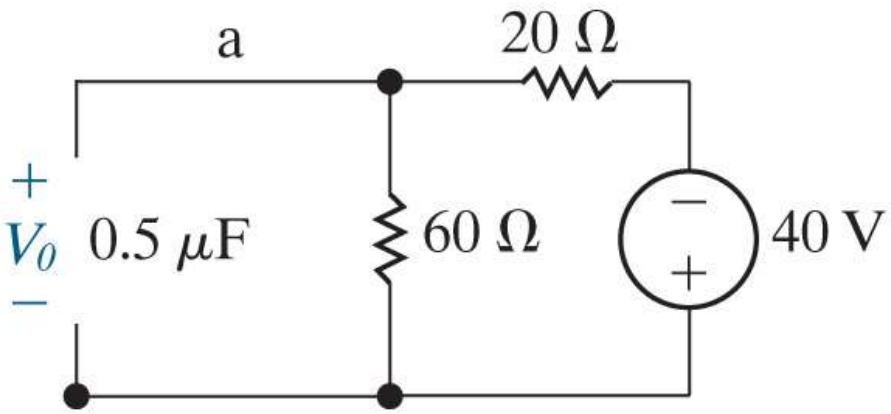
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Chapter 7 – *RL/RC* Step Response Examples (8 of 10)

What is the best circuit analysis tool to use when finding the initial voltage drop across the capacitor?

For $t < 0$



✗ A. Current division

✓ B. Voltage division

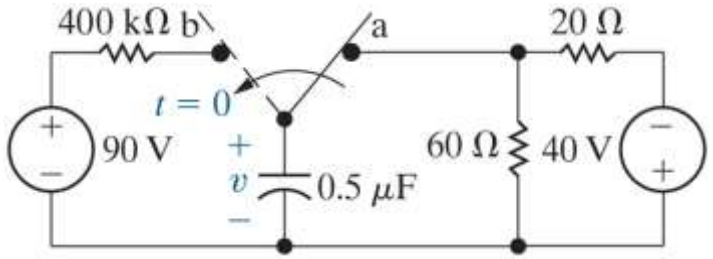
✗ C. Source transform

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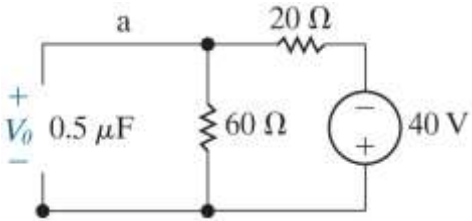
Chapter 7 – *RL*/*RC* Step Response Examples (9 of 10)

Find *v(t)* for $t \geq 0$ (continued).



Step 2 – Calculate the initial value of the voltage drop across the capacitor, V_0.

For $t < 0$

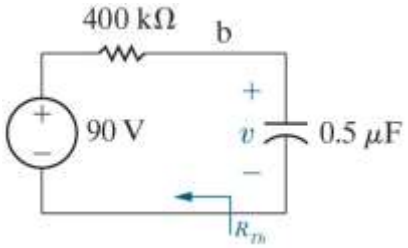


$$V_o = \frac{60}{60+20}(-40)$$

$$= -30 \text{ V}$$

Step 3 – Calculate the time constant $\tau = RC$.

For $t \geq 0$



$$R_{Th} = 400 \text{ k}\Omega$$

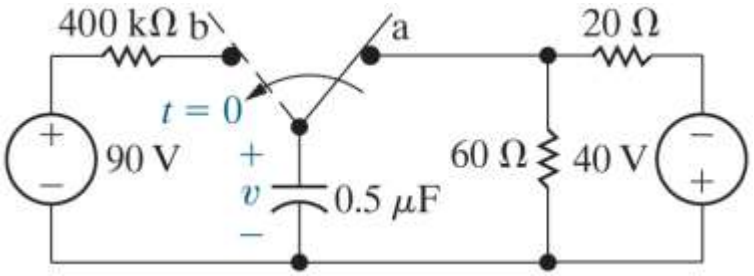
$$\tau = (400{,}000)(0.5 \times 10^{-6}) = 0.2 \text{ s}$$

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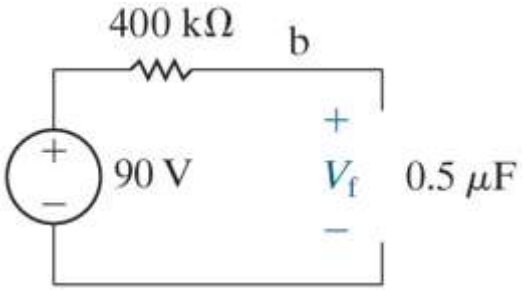
Chapter 7 – *RL*/*RC* Step Response Examples (10 of 10)

Find $v(t)$ for $t \geq 0$ (continued).



Step 4 – Calculate the final value V_f.

As $t \to \infty$



$$V_f = 90\,\text{V}$$

Step 5 – Write the equation for $v(t)$.

$$v(t) = V_f + (V_0 - V_f)e^{-t/\tau} = 90 + [-30 - 90]e^{-t/0.2}$$

$$= 90 - 120e^{-5t}\ \text{V}, \quad t \geq 0 \qquad \text{(check at } t = 0 \text{ and } t \to \infty\text{)}$$

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Chapter 7 – *RL*/*RC* Natural and Step Response (1 of 2)

This is a general solution method for both the natural and step responses of first-order circuits. A circuit exhibiting a natural response has no independent source as $t \to \infty$, so its final value $X_f = 0$.

Step 1 – Identify the variable $x(t)$, which is the inductor current for *RL* circuits and capacitor voltage for *RC* circuits.

Step 2 – Calculate the initial value X_0 by analyzing the circuit to find $x(t)$ for $t < 0$.

Step 3 – Calculate the time constant τ; for *RL* circuits, $\tau = R / L$ and for *RC* circuits, $\tau = RC$, where R is the equivalent resistance connected to the inductor or capacitor for $t \geq 0$.

Step 4 – Calculate the final value X_f by analyzing the circuit to find $x(t)$ as $t \to \infty$.

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Chapter 7 – *RL*/*RC* Natural and Step Response (2 of 2)

Step 5 – Write the equation for $x(t)$,

$$x(t) = X_f + (X_0 - X_f)e^{-t/\tau}, \quad \text{for} \quad t \geq 0.$$

Step 6 – Calculate other quantities of interest using $x(t)$.

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